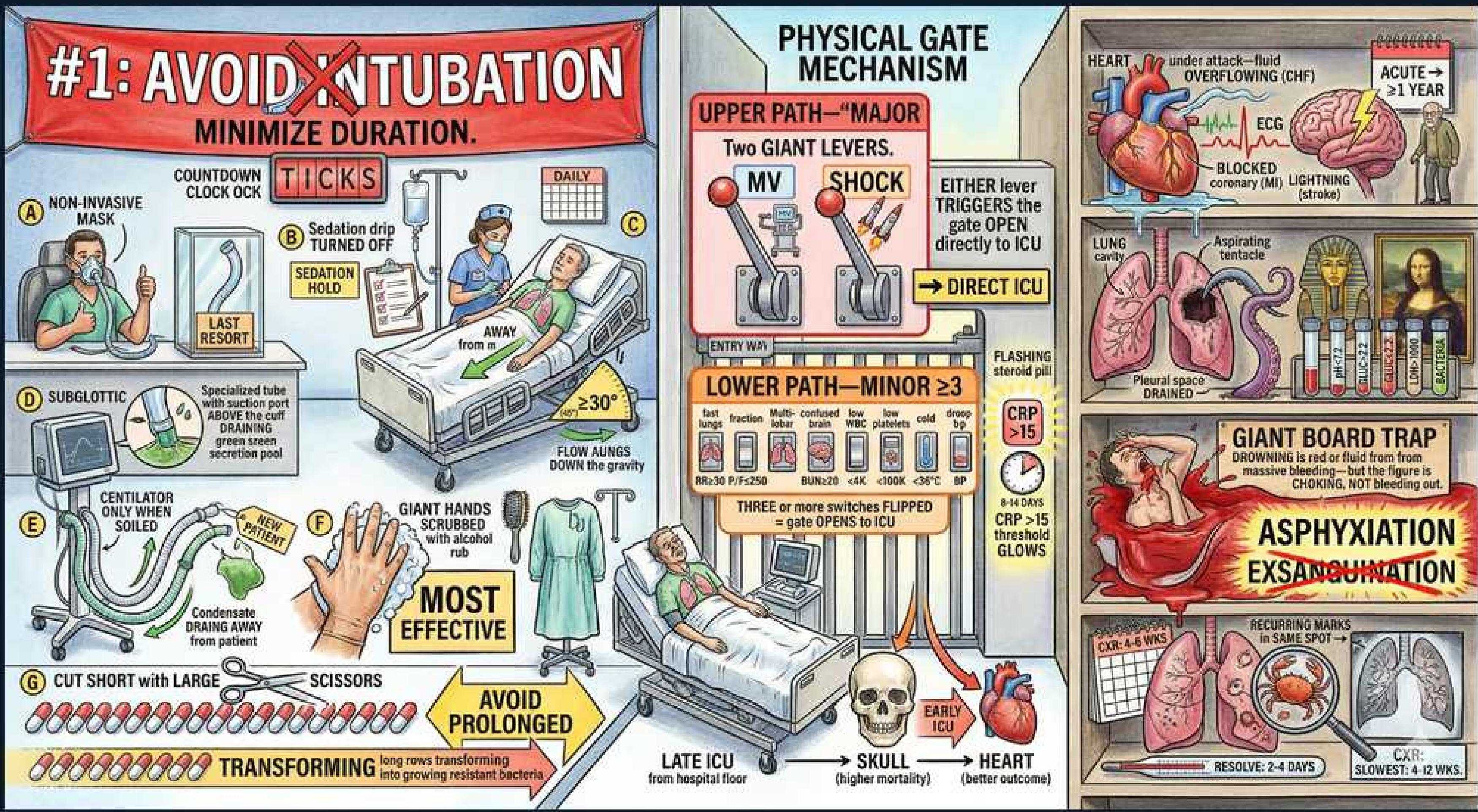
VAP Prevention Bundle & Severity / ICU Triage



1. Avoid intubation / NIV first

WHAT YOU SEE
Red banner '#1 AVOID INTUBATION', NIPPV mask

WHAT IT ENCODES
The single best VAP prevention is avoiding intubation: use non-invasive ventilation (NIV/HFNC) when appropriate. No endotracheal tube means no ventilator-associated pneumonia. Minimize duration when intubation is unavoidable.

BOARD TRAP
NIV is preferred in COPD exacerbation and cardiogenic edema but is risky in altered mental status or copious secretions — don't apply it to a patient who can't protect the airway.

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2. Minimize ventilation duration

WHAT YOU SEE
Countdown clock, daily spontaneous breathing trial

WHAT IT ENCODES
Daily sedation interruption ('sedation holiday') and spontaneous breathing trials shorten ventilator days and reduce VAP. Less time intubated = lower cumulative risk.

BOARD TRAP
Bundle elements reduce VAP incidence; over-sedation prolongs ventilation and raises risk — daily awakening trials are evidence-based.

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3. Subglottic suction tube

WHAT YOU SEE
'SUBGLOTTIC' specialized tube with suction port

WHAT IT ENCODES
Endotracheal tubes with subglottic secretion drainage remove pooled secretions above the cuff, reducing micro-aspiration and VAP. Part of the prevention bundle.

BOARD TRAP
Subglottic drainage targets secretions pooling above the cuff — the route of VAP — distinct from oral care, though both are bundle components.

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4. Head of bed elevation 30-45

WHAT YOU SEE
'AWAY', bed angle >=30 degrees

WHAT IT ENCODES
Semi-recumbent positioning (head of bed 30–45 degrees) reduces gastroesophageal reflux and aspiration of contaminated secretions, lowering VAP. Simple, no-cost bundle measure.

BOARD TRAP
Supine positioning increases aspiration risk; HOB elevation is a core VAP bundle item — flat positioning is a preventable error.

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5. Hand hygiene

WHAT YOU SEE
Giant scrubbed hands 'MOST EFFECTIVE'

WHAT IT ENCODES
Hand hygiene is the most effective single measure to prevent cross-transmission of nosocomial pathogens in the ICU. Reduces MDR colonization and infection.

BOARD TRAP
Hand hygiene prevents transmission of resistant organisms broadly — the highest-yield infection control intervention across all HAIs.

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6. Avoid condensate / circuit changes

WHAT YOU SEE
Condensate draining away, 'AVOID PROLONGED'

WHAT IT ENCODES
Avoid pooling ventilator-circuit condensate draining toward the patient and do not change circuits routinely (only when soiled), as frequent changes increase contamination risk.

BOARD TRAP
Routine circuit changes do NOT reduce VAP and may increase it — change only when visibly soiled/malfunctioning.

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7. Sedation hold / 'last resort'

WHAT YOU SEE
'SEDATION HOLD', 'LAST RESORT' tag

WHAT IT ENCODES
Daily sedation interruption assesses readiness to extubate and reduces ventilator days. Deep continuous sedation is a last resort. Paired with spontaneous breathing trials for liberation.

BOARD TRAP
Linking sedation holds to breathing trials (the 'ABCDE bundle') is what reduces VAP and delirium — sedation alone is not protective.

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8. Two major criteria → direct ICU

WHAT YOU SEE
'MAJOR' two giant levers: MV, shock → ICU

WHAT IT ENCODES
IDSA/ATS major severity criteria: septic shock requiring vasopressors OR respiratory failure requiring mechanical ventilation — either one mandates ICU admission for severe CAP.

BOARD TRAP
A single MAJOR criterion (shock or need for mechanical ventilation) sends the patient straight to ICU — don't wait to tally minor criteria.

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9. Three or more minor criteria

WHAT YOU SEE

'MINOR >=3' lower levers, switches flipped

WHAT IT ENCODES

Severe CAP is also defined by >=3 minor IDSA/ATS criteria (e.g., RR>=30, PaO2/FiO2<=250, multilobar infiltrates, confusion, uremia, leukopenia, thrombocytopenia, hypothermia, hypotension needing fluids). Three or more → consider ICU.

BOARD TRAP

You need at least 3 minor criteria (not 1–2) to qualify as severe CAP by the minor-criteria pathway.

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10. CRP/biomarker threshold

WHAT YOU SEE

'CRP >15' glowing meter

WHAT IT ENCODES

Inflammatory markers (CRP, procalcitonin) support severity assessment and can guide antibiotic duration, but are adjuncts to clinical scoring, not standalone ICU triggers.

BOARD TRAP

Biomarkers refine but never replace clinical/physiologic severity criteria — don't admit to ICU on CRP alone.

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11. CURB-65 / mortality scoring

WHAT YOU SEE

'higher mortality', skull vs heart panel

WHAT IT ENCODES

CURB-65 (Confusion, Urea>7, RR>=30, BP<90/60, age>=65) stratifies CAP mortality and disposition; PSI/PORT is the alternative. Higher scores → inpatient/ICU care.

BOARD TRAP

CURB-65 guides admission decisions but the IDSA/ATS major/minor criteria specifically guide ICU admission — different tools for different decisions.

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12. CHF / cardiac complication

WHAT YOU SEE

Overflowing heart 'acute >1 yr', CHF

WHAT IT ENCODES

Pneumonia precipitates cardiac events (new/worsening heart failure, arrhythmia, MI) — a major driver of post-pneumonia mortality, with elevated risk persisting beyond a year.

BOARD TRAP

Cardiovascular complications, not just respiratory failure, drive late pneumonia mortality — monitor for MI/CHF during and after the illness.

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13. Empyema / drainage

WHAT YOU SEE

Pleural space drained, fluid

WHAT IT ENCODES

Complicated parapneumonic effusion or empyema (pus, low pH<7.2, loculations) requires chest-tube drainage in addition to antibiotics. Failure to drain prolongs sepsis.

BOARD TRAP

A parapneumonic effusion with pH<7.2, glucose<60, or frank pus needs drainage — antibiotics alone will fail an empyema.

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14. Asphyxiation vs exsanguination trap

WHAT YOU SEE

Giant board trap: 'ASPHYXIATION not EXSANGUINATION'

WHAT IT ENCODES

In massive hemoptysis the cause of death is asphyxiation (airway flooding/drowning), not blood loss. Position the bleeding lung down, protect the airway, and intubate to the good lung.

BOARD TRAP

Classic trap: patients with massive hemoptysis die from asphyxiation, not exsanguination — airway protection is the priority over transfusion.

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15. Resolution timeline

WHAT YOU SEE

'RESOLVE 2-4 DAYS', recurring same spot warning

WHAT IT ENCODES

Clinical improvement (fever, WBC) is expected within ~48–72h of effective antibiotics; CXR clears more slowly. Pneumonia recurring in the same location should prompt evaluation for obstruction/malignancy.

BOARD TRAP

Non-resolving or same-site recurrent pneumonia warrants bronchoscopy/CT to exclude an obstructing lesion — don't just re-treat.
